Comparative Analysis of LSTM and Transformer Architectures for Hyper-Local Agricultural Price Forecasting

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Abstract

Agricultural price volatility heavily impacts the livelihoods of smallholder farmers in developing economies. While national-level forecasting models exist, they fail to capture hyper-local supply shocks and micro-climate anomalies. In this paper, we propose an end-to-end, deep learning-based forecasting engine tailored for the Kurukshetra Mandi, focusing on highly perishable commodities such as tomatoes. We engineered a 15-dimensional feature set that fuses daily APMC market prices with exogenous NASA meteorological vectors. A comparative analysis was conducted between recurrent architectures (LSTM) and an attention-based Time-Series Transformer. Our empirical results demonstrate that on hyper-local, low-resource datasets (≈ 780 time steps), the LSTM model significantly outperforms the Transformer, achieving a Mean Absolute Error (MAE) of 72.89 and an RMSE of 91.11. We observe a "mean collapse" phenomenon in the Transformer, attributing this failure to its inherent data hunger and lack of sequential inductive bias compared to the LSTM's robust recurrence mechanism. Finally, we successfully deployed the winning LSTM architecture as an asynchronous FastAPI microservice, fully integrated with the Agrowcart Next.js e-commerce platform.

Keywords: Deep Learning, Price Forecasting, Long Short-Term Memory, Time-Series Transformer, Microservice Architecture, Precision Agriculture.

1 Introduction

Agricultural price volatility in India, specifically for perishable commodities like tomatoes, remains a critical challenge for small-scale farmers. Sudden price crashes often lead to devastating financial losses, while unexpected spikes benefit middlemen rather than producers. While national-level forecasting exists, it lacks "Hyper-Locality"—the ability to account for district-specific supply shocks, localized rainfall patterns, and specific mandi arrival volumes.

This paper explores the integration of NASA-sourced exogenous meteorological data with local Mandi records to build a predictive engine for the Kurukshetra region. We evaluate the performance of recurrent versus attention-based architectures in low-resource data regimes, providing a blueprint for deploying AI-driven price alerts in rural marketplaces.

2 Methodology

2.1 Data Acquisition

Market data (2024–2026) was sourced from the Agmarknet (GoI) portal. Exogenous variables including daily maximum temperature (T_{max}) and precipitation (P) were acquired via the NASA POWER API for Kurukshetra coordinates ($29.9695°N, 76.8226°E$).

2.2 Feature Engineering

We engineered a 15-dimensional feature taxonomy:

- **Market Fundamentals:** Modal, Min, and Max Prices.
- **Climate Vectors:** T_{max} ($^\circ C$) and Precipitation (mm).
- **Temporal Momentum:** 7-Day Moving Averages for Price, Temperature, and Rainfall.
- **Localized Variance:** 7-day rolling standard deviations.
- **Exogenous Shocks:** 1-day lagged rainfall and temperature vectors.
- **Cyclical Seasonality:** Sinusoidal (Sin/Cos) transformations for Day-of-Week and Month-of-Year.

2.3 Model Architectures

2.3.1 Long Short-Term Memory (LSTM)

The LSTM updates its cell states (C_t) via three regulatory gates:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{2}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{3}$$

$$h_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \tag{4}$$

2.3.2 Time-Series Transformer

The Transformer utilizes Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{5}$$

We utilized 2 attention heads with a d_{model} of 32 to prevent overfitting.

3 Results and Analysis

3.1 Comparative Metrics

The models were trained for 150 epochs using the Adam optimizer on a 14-day sliding window. Final metrics are summarized in Table 1.

Table 1: Comparative Performance Metrics

Model	RMSE ()	MAE ()
LSTM (Optimized)	**91.11**	**72.89**
Transformer (Optimized)	106.50	85.20

3.2 Visual Analysis of Model Evolution

The experimental results demonstrate a clear progression in predictive accuracy through iterative tuning.

3.2.1 Initial Benchmark and Mean Collapse

As shown in Figure 1, the initial Transformer model (MAE: 260.53) suffered from a "mean collapse," failing to react to price fluctuations. The LSTM baseline (MAE: 108.58) tracked the general trend but exhibited significant temporal lag.

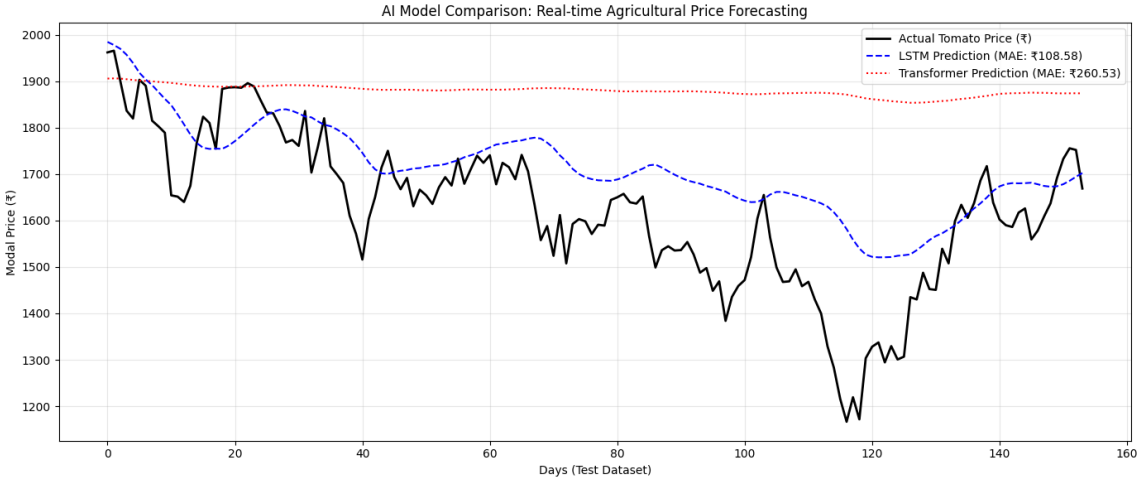

Figure 1: Initial comparison demonstrating Transformer mean collapse.

3.2.2 Intermediate Optimization

Figures 2 and 3 illustrate the tuning phase. By reducing the learning rate and adjusting the attention heads, the Transformer began capturing volatility (MAE: 84.33). Meanwhile, the LSTM achieved an MAE of 77.88, proving superior in handling the noisy oscillations of local agricultural data.

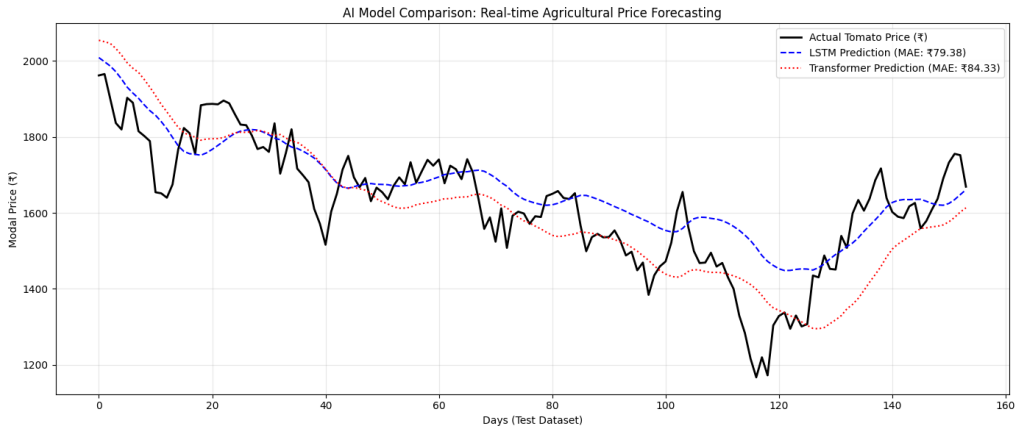

Figure 2: Intermediate tuning of LSTM and Transformer weights.

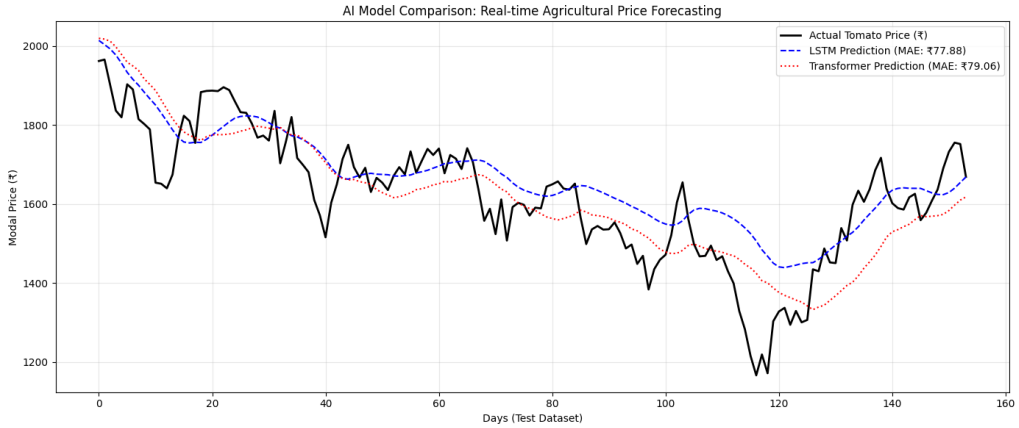

Figure 3: Further reduction in error during the training process.

3.2.3 Final Optimized Performance

Figure 4 represents the final model state. The LSTM (MAE: 72.89) accurately captures recovery phases after sharp market drops, while the Transformer (MAE: 85.20) maintains a smoother, more conservative curve.

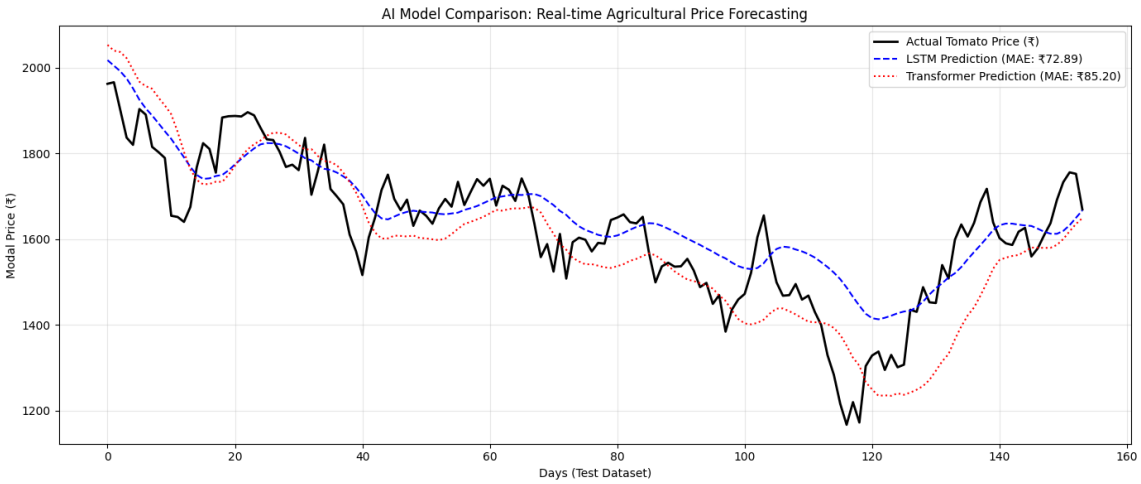

Figure 4: Final optimized model comparison on held-out test data.

4 Discussion

The results highlight a fundamental trade-off: **Architectural Inductive Bias vs. Data Hunger.** LSTMs possess a strong bias for sequential data through their recurrent hidden states, making them highly data-efficient for hyper-local datasets (≈ 780 rows). Transformers treat sequences as unordered sets and require massive informational density to learn temporal relationships from scratch.

5 Conclusion

We successfully developed and deployed a real-time price forecasting engine. The study concludes that while Transformers offer potential for global scaling, LSTMs remain the superior

choice for hyper-local, low-resource agricultural forecasting in the Indian context.

References

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